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A New Discrete Learning-Based Logistic Regression Classifier for Bankruptcy Prediction

Mehdi Khashei[1,2] · Sepideh Etemadi[1] · Negar Bakhtiarvand[1]

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Abstract
Credit scoring or predicting bankruptcy is among the most crucial techniques for identifying high-risk and low-risk credit situations. Accordingly, enhancing the accuracy of bankruptcy prediction methods decreases the risk of inappropriate financial decisions. Also, increasing the accuracy of credit scoring models brings significant benefits such as improved turnover, credit market growth, proper and efficient allocation of financial resources, and sustained improvement of the profits of banks, investors, funds, and governments. Various statistical classification methods have been developed in the literature with different features and characteristics for more accurate bankruptcy prediction. However, despite all appearance differences in statistical classification approaches, they all adhere to a common idea and concept in their training procedures. The basic operation logic in whole-developed statistical classification methods focuses on maximizing a continuous distance-based cost function to yield the highest performance. Despite it being a common and frequently used procedure for classification purposes, it is an unreasonable and inefficient manner to achieve maximum accuracy in a discrete classification field. In this paper, a new discrete direction-based Logistic Regression that is a common statistical classifier method for bankruptcy forecasting is proposed. In the proposed Logistic Regression, in contrast to all traditionally developed statistical classifiers, the compatibility of the cost function and the training procedure is considered. While it can be shown overall that the performance of the presented discrete direction-based classifier will not be inferior to its continuous counterpart, an evaluation of the suggested classifier is conducted to ascertain its superiority. For this purpose, three credit scoring datasets are considered to assess the classification rate of the presented classifier. Empirical outcomes demonstrate that, as pre-expected, in all cases, the model put forward can attain a superior performance compared to conventional alternatives. These findings clearly demonstrated the significant influence of the consistency between the cost function and the training process on the classification capability, a consideration absent in any of the traditional statistical classification procedures. Consequently, the presented Logistic Regression can be considered an efficient alternative for credit scoring purposes to achieve more accurate results.

Keywords Discrete and continuous learning algorithms · Statistical classification procedures · Logistic regression · Classification rate · Credit scoring · Bankruptcy prediction

Extended author information available on the last page of the article

1 Introduction

Predicting bankruptcy stands out as a pivotal technique for identifying high-risk and low-risk credit situations. Accordingly, enhancing the accuracy of bankruptcy prediction models brings significant benefits such as improved turnover, proper and efficient allocation of financial resources, credit market growth, and sustained improvement of the profits of investors, banks, funds, and governments. A variety of statistical data mining methods with different characteristics such as Logistic Regression (LR), Probit Regression (PR), Multivariate Discriminant Analysis (MDA), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Naïve Bayes (NB), Bayesian Network (BN), K-Nearest Neighbor (KNN), Gaussian Process Classifier (GPC), etc., have been developed for credit scoring purposes [1–4]. Among statistical approaches, Logistic Regression is a prevalent statistical classification technique widely employed in data mining and modeling. However, all classifier models in the credit scoring applications used a continuous distance-based function as a learning cost function for a discrete classification purpose. Silva et al. [5] established an LR approach based on the covariate categorization concept for loan default forecasting. Ahelegbey et al. [6] constructed an LR classifier based on the latent factor-based classification method for corporate default forecasting.

Filippopoulou et al. [7] created an LR to forecast banking crises in the Eurozone. This model could be used as guidance in establishing financial policies. Lee and Jun [8] established the method for learning the LR classifier in the tree for financial applications. Klieštik et al. [9] applied LR and Penalized Logistic Regression (PLR) to predict the financial distress of firms. Dumitrescu et al. [10] suggested Penalized Logistic Tree Regression (PLTR) by incorporating feature selection derived from the Decision Tree (DT) into the LR in order to model nonlinear structures. The presented method outperformed LR, Random Forest (RF), Nonlinear LR (NLR) with the adaptive lasso, Artificial Neural Network (ANN), and Support Vector Machine (SVM) in the Australian credit scoring data set. Villuendas-Rey et al. [11] presented a Naive Associative Classifier (NAC) model for various finance applications such as bankruptcy, bank telemarketing, banknote authentication, and credit assignment. Sohn et al. [12] developed fuzzy LR for evaluating credit ranking in technology-based corporations. Nikolic et al. [13] established the brute force LR model by considering feature selection techniques for assessing credit risk in Serbian companies. Vrontos et al. [14] compared and assessed the performances of KNN, PLR, and Bayesian Generalized Linear (BGL) methods to forecast U.S. recession probability.

Teply and Polena [15] assessed the performance of various bankruptcy prediction models. The empirical results demonstrated the superiority of LR, LDA, and ANN classifiers against other methods. Moscatelli et al. [16] compared the classification accuracy of LR, PLR, LDA, RF, and Gradient Boosting Tree (GBT) models to determine the creditworthiness of borrowers. Wang et al. [17] compared and assessed the discriminant power of LR, NB, KNN, RF, and DT classifiers for detecting default and solvent applicants. Barboza et al. [18] evaluated the accuracy of several classification methods, such as RF, SVM, and ANN, against statistical methods like LR and MDA for bankruptcy forecasting in Canadian and American corporations. Fitzpatrick and Mues [19] assessed the classification rate of PLR and semi-parametric LR, RF, and Boosted Regression Trees (BRT) for diagnosing loan default customers. Abdou et al. [20] compared the prediction power of LR, Cascade Correlation Neural Network (CCNN), and Classification and Regression Tree (CART) approaches for credit ranking. Gordini [21] evaluated the accuracy of the LR and SVM classifiers for bankruptcy forecasting in Italian

manufacturing corporations. Cubiles-De-La-Vega et al. [22] used several classifiers such as LR, QDA, SVM, Multilayer Perceptrons (MLP), classification trees, and hybrid structures for assessing the creditworthiness of corporates. Lin [23] investigated the accuracy of LR, PLR, MDA, and ANN classifiers for loan default prediction in Taiwan.

Despite all the efforts made in the development of classification approaches to achieve the maximum classification rate, the use of a continuous distance-based training procedure for training and determining the parameters of the model is the common point of all conventional approaches. On the other hand, classification methods stand out as one of the extensively employed data mining techniques in various fields of science for the purposes of diagnosis, discrimination, and detection, which fall into the category of discrete problems. Hence, proposing a novel classification approach that utilizes a discrete direction-based cost function for the training process and determining the parameters of the classifier, in order to match the learning process with the objectives of classification is an effective and sensible strategy. It represents an outstanding method compared to traditional classification methods for attaining the highest level of performance. Thus, by substituting the typical learning procedure with the proposed learning process in the Logistic Regression classifier—a highly utilized and favored statistical model— a new version of discrete learning-based LR is developed. It should be highlighted that while the suggested procedure is applicable to all statistical/intelligent classifiers, this paper specifically implements the proposed methodology on Logistic Regression. Overall, we were motivated to select Logistic Regression over alternative statistical classifiers to implement the proposed methodology for three reasons: 1) LR possesses the simplest overall formulation process, 2) LR has the lowest number of parameters/hyperparameters, and 3) LR is one of the most favored and extensively utilized statistical classification models in the literature.

Furthermore, the assessment of the effectiveness and superiority of the proposed discrete direction-based logistic regression (DILR) over all state-of-the-art classifier methods is shown, with a focus on the classification rate. This evaluation is specifically conducted in the field of bankruptcy prediction using three widely-used benchmark datasets, namely Australian, German, and Japanese. While it can be demonstrated overall that the performance of the presented DILR classifier will not be inferior to its traditional continuous distance-based counterpart, in this manner, the principal hypotheses/questions of the study can be stated as follows:

1) Is the suggested discrete cost function a suitable replacement for the continuous counterpart?
2) Is it possible for the suggested discrete learning-based algorithm to exhibit greater efficiency compared to its traditional continuous one?
3) Could the suggested Logistic Regression produce results of higher accuracy than its traditional version?

Additionally, the principal novelties of this research can be outlined as follows:

- This study introduces an innovative learning approach for classification issues, employing a discrete direction-based matching function. This is unlike the training procedure of conventional classifiers that rely on typical continuous distance-based cost functions.
- The core notion of the suggested approach is adjusting the training procedure of classifiers with the discrete form of classification.

- The presented approach is applied to determine the parameters of logistic regression, which is recognized as one of the most frequently employed statistical models.
- The effectiveness of the suggested discrete direction-based logistic regression is examined using three benchmark data sets in the credit scoring domain, and the outcomes are compared with the conventional counterpart as well as various commonly employed and recently developed methods.

The subsequent parts of the study are designed as follows. Section 2 introduces the mathematical formulation of the Logistic Regression, utilizing the concept of the discrete direction-based learning process. Section 3 outlines the chosen datasets for assessing the proposed model and introduces the evaluation metrics employed for the model. Additionally, Sect. 4 includes an examination and comparison of the presented classifier's performances with conventional methods in the field of bankruptcy prediction. Ultimately, conclusions and future research opportunities are expressed.

2 The Discrete Learning-based Logistic Regression (DILR)

Logistic regression stands out as one of the frequently employed statistical algorithms for addressing classification problems [24]. Several research has been conducted to enhance the performance of logistic regression-based models. In general, a k-variable LR classifier encompasses the target variable Y and $k-1$ independent variables $X_2, X_3, ..., X_k$ can be overall written as follows:

$$\begin{aligned} P_t &= E(Y_t) = \frac{e^{\beta_1+\beta_2 X_{2i}+\beta_3 X_{3i}+\ldots+\beta_k X_{ki}}}{1+e^{\beta_1+\beta_2 X_{2i}+\beta_3 X_{3i}+\ldots+\beta_k X_{ki}}} && t = 1, 2, \ldots, N \\ logit(P_t) &= Ln\left(\frac{P_t}{1-P_t}\right) = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \ldots + \beta_k X_{ki} && t = 1, 2, \ldots, N \end{aligned} \tag{1}$$

where, β_1 is the intercept, $\beta_2, ..., \beta_k$ are partial slope parameters, and N is the sample size. The key assumptions underlying classic LR are as follows: **1)** the independent variables exhibit a linear relationship with the log of odds; **2)** the dependent variable is binary; **3)** the observations are independent of each other; **4)** there is little or no multicollinearity among the explanatory variables. The Maximum Likelihood (ML) method is one of the most significant and well-known statistical approaches often used by statisticians to estimate unknown coefficients in the LR model. While this cost function is widely used, it takes on a continuous form, whereas the classification goal function is discrete. Consequently, a notable challenge in this type of learning process stems from the irrationality or, at the very least, inefficiency of employing a continuous distance-based cost function in discrete classification domains. The rationale behind this lies in the use of a continuous learning-based cost function in training processes. In these processes, the fitted values progressively approach the actual value during each training epoch. To put it differently, the distance between the estimated and actual amounts consistently decreases in each epoch of the training procedure. This approach is particularly effective when dealing with problems in the continuous modeling class, including continuous time series analysis or causal forecasting. In such cases, there is a general assurance that the model's performance will continuously enhance. Nevertheless, this assertion does not overall hold true in the discrete modeling class, including discrete time series analysis or classification. This is due to the fact that the ultimate outcomes of a continuous training procedure within discrete modeling must undergo discretization prior to being presented as the predicted amount of the

target variable. As a result, if the directions of the "training" and "discretization" phases align, the training procedure may contribute to performance improvement. Conversely, if the direction of the "training procedure" contradicts that of the " discretization procedure" the impact of the former may be nullified by the latter. Thus, it can be inferred that employing a continuous learning-based procedure is irrational and may be entirely unsuitable and highly inefficient. Thus, this study introduces a discrete learning-based method for statistical classification to yield a more reasonable algorithm. In this manner, the proposed discrete-learning based method for determining the unknown weights of the LR classifier can be formulated as follows:

$$Max \sum_{t=1}^{N} Match\left(y_t, \hat{y}_t\right) \tag{2}$$

where, the $Match\left(y_t, \hat{y}_t\right)$ is the matching function of the actual ($y_t \in \{-1, +1\}$) and the estimated ($\hat{y}_t \in \Re$) at time t. As can be seen, the proposed discrete matching function for determining the unknown weights of the LR classifier tries to maximize the number of matching cases, cases in which the sign of the estimated amount matches with the sign of the actual one, in the whole of the training samples. In this way, the binary form of the proposed matching function can be mathematically represented as follows:

$$Match\left(y_t, \hat{y}_t\right) = \begin{cases} +1 \ if \ \left(y_t\right)\left(\hat{y}_t^s\right) \geq 0 \\ -1 \ if \ \left(y_t\right)\left(\hat{y}_t^s\right) < 0 \end{cases} \tag{3}$$

where, $\hat{y}_t^s = Std.\left(\hat{y}_t\right)$ is the standardized amount of the $\hat{y}_t$ at time t that is computed as Eq. (4). As can be seen, in the proposed discrete matching function, when the sign of the estimated amount matches with the sign of the actual value, or mathematically equivalently, when the sign of the multiplication of the estimated value in the actual value is positive or zero; then, the assigned value by the matching function will be equal to "+1". In contrast, when the sign of the estimated value does not match with the sign of the actual one, or mathematically equivalently when the sign of the multiplication of the estimated value in the actual value is negative; then, the assigned value by the matching function will be equal to "-1". In addition, in this function, the standard value of the estimated value is instead used of the pure estimated value. From the theoretical point of view, this action will be ineffectual, due to the fact that the sign of a value will be equal before and after of standardization process. However, from a practical perspective, it can reduce the variance of the fitted amounts, which can lead the proposed model to work faster.

$$\hat{y}_t^S = \frac{\hat{y}_t - \bar{\hat{y}}}{\hat{y}^{Max} - \hat{y}^{Min}} \tag{4}$$

The $\bar{\hat{y}}_t$,$\hat{y}^{Min}$, and $\hat{y}^{Max}$ are the Mean, Minimum, and maximum of $\hat{y}_t$. Now, Eq. (2) can be presented by the *Sign* function as follows:

$$Max \sum_{t=1}^{N} Sign\left[\left(y_t\right).\left(\hat{y}_t^s\right)\right] \tag{5}$$

Or, in a more simplified form:

$$Max \sum_{t=1}^{N} Sign\left(y_t\right).Sign\left(\hat{y}_t^s\right) \tag{6}$$

Therefore, we have the following:

$$Max \sum_{t=1}^{N} \left(y_t\right).Sign\left(Std.\left(Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \Big/ 1 + Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \right) \right) \tag{7}$$

Finally, as for the linearization of the proposed algorithm Eq. (7) can be converted to a mixed-integer programming form; in which,ε and M and are a very small and a very large number; respectively, as follows:

$$Max \sum_{t=1}^{N} Sign\left(y_t\right).Sign\left(\hat{y}_t^s\right) = \sum_{t=1}^{N} \left(y_t\right).Sign\left(Std.\left(Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \Big/ 1 + Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \right) \right)$$

$$S.T \begin{cases} Std.\left(\left(Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \Big/ 1 + Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \right) \right) \geq \varepsilon - M(1 - a_t) & t = 1,2,3,...,N \\ Std.\left(\left(Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \Big/ 1 + Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \right) \right) \leq -\varepsilon + M(1 - b_t) & t = 1,2,3,...,N \\ Std.\left(\left(Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \Big/ 1 + Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \right) \right) \geq -M(1 - c_t) & t = 1,2,3,...,N \\ Std.\left(\left(Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \Big/ 1 + Exp\left(\sum_{k=0}^{k} \hat{\beta}_k.X_{t,k} \right) \right) \right) \leq M(1 - c_t) & t = 1,2,3,...,N \\ a_t + b_t + c_t = 1 & t = 1,2,3,...,N \\ Sign\left(\hat{y}_t^s\right) = a_t - b_t & t = 1,2,3,...,N \\ a_t, b_t, c_t \in \{0,1\} & t = 1,2,3,...,N \\ \hat{\beta}_j & free\ of\ sign \end{cases} \tag{8}$$

As can be seen from Eq. (8), the complexity of the proposed DILR classifier is fundamentally dependent on the number of data, rather than the number of predictor features. It comes from the fact that the number of integer parameters has a direct relationship with the sample size. In other words, the proposed model needs three integer parameters for each data. Ultimately, the proposed DILR model's key strengths and weaknesses are outlined. Generally, the literature introduces various features essential for the practical assessment and comparison of diverse modeling methods. Among the most significant considerations, ranked by significance, are accuracy, computational time and cost, interpretability, and implementation and ease of use. Numerical findings will reveal that the DILR is capable of delivering more accurate outcomes compared to traditional LR and some other single/hybrid statistical/intelligent methods. Hence, the significant strength of the presented classifier, distinguishing it from other statistical methods, lies in its accuracy. However, the computational cost and time associated with the DILR classifier, attributable to its discrete variables, exceed those of its traditional counterpart. Therefore, the most significant drawback of the presented classifier is its computational cost and time. Additionally, like traditional LR and other statistical classifiers, the proposed model boasts high interpretability. Consequently, the second strength of the presented classifier lies in its high interpretability. Similarly, the process for designing and specifying the desired structure of the presented classifier, resembling traditional LR and

some other statistical models, is straightforward. Thus, the third advantage is that utilizing and implementing the DILR classifier is not a problematic/challenging task.

3 Data Description and Evaluation Criteria

In this study, 3 credit scoring benchmark datasets namely Australian, German, and Japanese are regarded to comprehensively assess the proposed DILR classifier. The assessment and comparison of the performance between the proposed DILR and the traditional LR have been conducted with a focus on the classification rate. These datasets differ in sample size, types and numbers of explanatory variables, and balance rate. The sample size of credit scoring datasets varied from 653 to 1000 data points and the balance rate, the class rate of the positive class, is also changed from 30.00 to 55.51%. The explanatory variables in these datasets are categorical, real, or integer and their number is varied from 14 to 24 attributes. Additional details about these datasets, including attributes' characteristics, the number of features, the total sample size, and the balance rate, have been reported in Table 1.

4 Empirical Result

The dataset titled "Japanese credit scoring" is chosen as an instance, and the implementing process of the proposed DILR model on this dataset is outlined. This dataset includes information on 15 explanatory variables of 653 credit scoring observations, used to differentiate between solvent and insolvent applicants. The explanatory features are as follows: X_2, X_3, X_8, X_{11}, X_{14}, X_{15} (continuous); X_1, X_9, X_{10}, X_{12} (categorical: 0–1); X_4, X_5, X_{13} (categorical: 1–3); X_6 (categorical: 1–14); and X_7 (categorical: 1–9). It must be noted that no details have been reported in the literature for explanatory variables of this data set, and they have been entirely converted to meaningless symbols by the producer to protect the confidentiality of the data. Statistical features of the dataset are depicted in Table 2. In addition, a chart illustrating the relationship between the predictor features based on the target variable is presented in Fig. 1. Pair-plot of explanatory variables against each other indicates the level of linear separability, or equivalently, the level of nonlinearity of the underlying data set. Generally, more overlapping of two classes in the pair plot indicates that such classes cannot be easily separated by

Table 1 Detailed information regarding the characteristics of the credit scoring benchmark datasets

No	Name	Sample size	Number of explanatory variables	Class rate (1)–class rate (2) (balance rate)	Characteristics of attributes
1-	Australian	690	14	55.51%–44.49% (55.51%)	Integer, real, categorical
2-	German	1000	24	30.00%–70.00% (30.00%)	Integer, categorical
3-	Japanese	653	15	54.67%–45.33% (54.67%)	Integer, real, categorical

Table 2 The statistical features of the Japanese credit scoring dataset

Features	Mean	Standard deviation	Mode	Median	Minimum	Maximum	Skewness	Kurtosis
X_2	31.504	11.838	22.670	28.420	13.750	76.750	1.080	0.821
X_3	4.830	5.027	1.500	2.835	0	28	1.482	2.230
X_8	2.244	3.371	0	1	0	28.500	2.916	11.337
X_{11}	2.502	4.968	0	0	0	67	5.048	48.803
X_{14}	180.360	168.297	0	160	0	2000	2.735	21.406
X_{15}	1013.761	5253.279	0	5	0	100,000	13.370	218.811

a linear classifier. In this way, since both the proposed logistic regression as well as its conventional version are linear classifiers, it is generally expected that their overall performance in such situations will be lower than non-linear classifiers. As evident from Fig. 1, the "Japanese credit scoring", as expected from almost all financial data sets, has a high level of nonlinearity.

In this study, 522 data points, constituting 80% of the dataset, are randomly chosen as training data, while the remaining 131 data points, representing around 20% of the dataset, are designated as the test sample. Moreover, to remove the impact of the random selection of the data, the model estimation procedure is iterated 100 times. The modeling of both the DILR and LR classifiers in this paper is conducted using the GAMS and MATLAB software packages. In addition, this study employs five assessment measures to evaluate the efficacy of the presented classifier and contrasts it with the conventional logistic regression. The evaluation criteria encompass classification rate, recall, precision, F1 score, and Kappa. The computation for these evaluation criteria is detailed as follows [25]:

$$Classification\,Rate = \frac{True\,Negative + True\,Positive}{False\,Negative + True\,Negative + False\,Positive + True\,Positive} \tag{9}$$

$$Recall = \frac{True\,Positive}{False\,Negative + True\,Positive} \tag{10}$$

$$Precision = \frac{True\,Positive}{False\,Positive + True\,Positive} \tag{11}$$

$$F1\,Score = \frac{2(Recall \times Precision)}{Recall + Precision} = \frac{2(True\,Positive)}{2(True\,Positive) + (False\,Positive + False\,Negative)} \tag{12}$$

$$Kappa = \frac{2(TN \times TP) - (FP + FN)}{(FP + TP) \times (TN + FP) + (FN + TP) \times (TN + FN)} \tag{13}$$

In this context, False Positive (FP) indicates negative data mistakenly identified as positive, True Positive (TP) signifies positive data correctly classified as positive, False Negative (FN) refers to positive data mistakenly classified as negative, and True Negative (TN) represents negative data correctly classified as negative. The mentioned performance metrics for both the DILR and the LR model, along with the percentage improvement of the presented classifier over the traditional counterpart, are provided in Table 3.

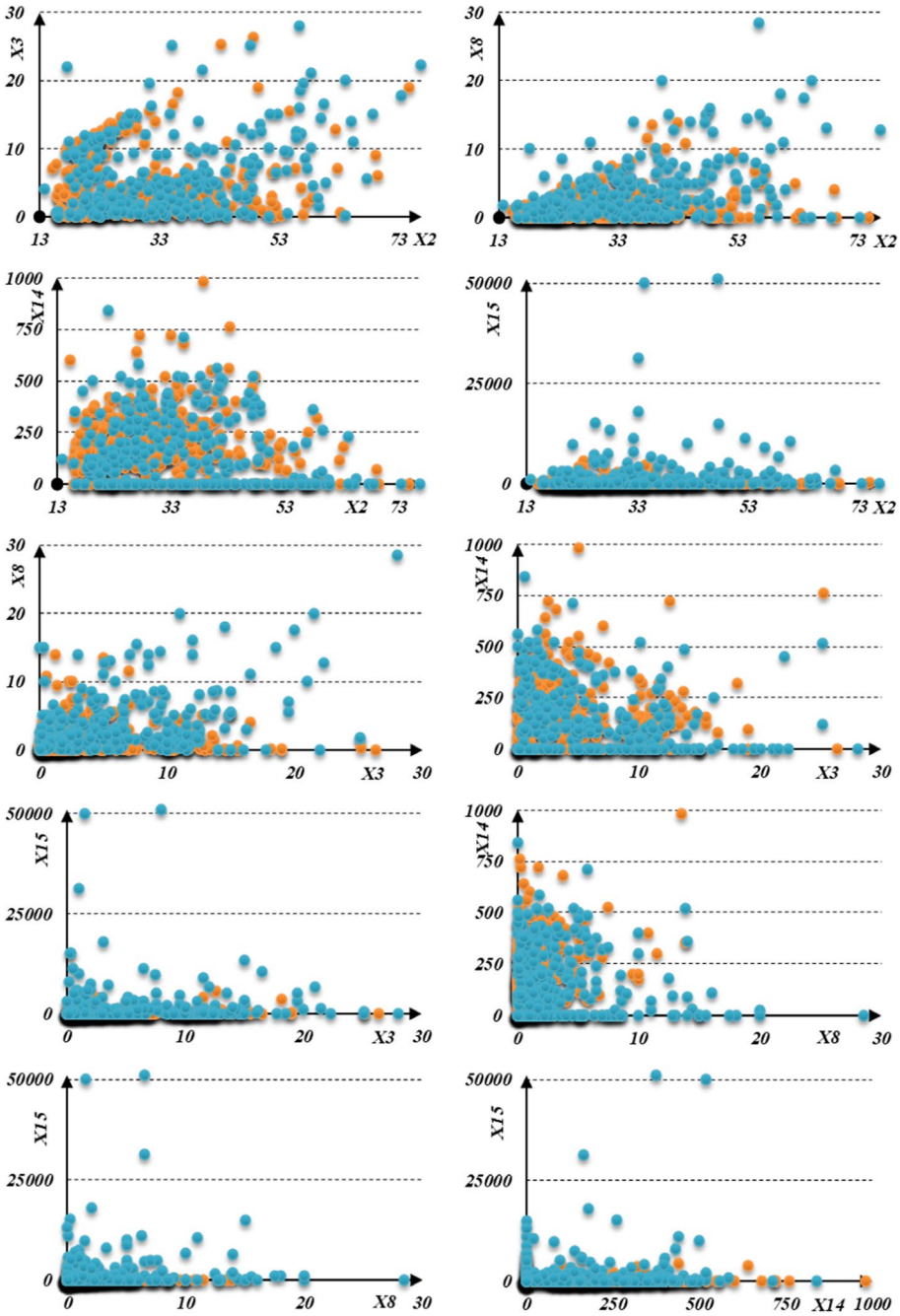

Fig. 1 Plot depicting the relationship between explanatory variables based on their target classes (blue: 0, orange: 1)

Table 3 Performance of the DILR and LR classifiers for the Japanese credit scoring dataset

Criteria	Japanese credit scoring data set		Improvement (%)
	Proposed DILR (%)	Classic LR (%)	
Classification rate	91.58	85.33	7.32
Precision	92.42	91.12	1.43
Recall	92.16	80.06	15.11
F1 score	92.29	85.23	8.28
Kappa	83.37	71.29	16.94
AUC	92.42	91.12	1.43

The performances reported in Table 3 indicate that the suggested DILR classifier can obtain more accurate outcomes compared to its traditional counterpart across all evaluation metrics. For example, the suggested DILR classifier can attain a 91.58% classification rate, which is 7.32% higher than classic logistic regression. Moreover, the proposed classifier can improve 1.43%, 15.11%, and 8.28% of the performance of the traditional LR in the precision, recall, and F1 score, respectively. Moreover, the DILR classifier can obtain a performance of 83.37% in the Kappa criterion, while the classic logistic regression can only reach 71.29% in this evaluation metric. It shows that the suggested classifier can enhance about 17% of the performance of the classic logistic regression in the Kappa criterion. The notably superior performance of the DILR classifier compared to the traditional counterpart in the Japanese benchmark dataset emphasizes the significance and effectiveness of the compatibility between the discrete learning-based procedure and the classification cost function. Furthermore, to conduct a more extensive assessment of the presented discrete learning-based classifier, its performance is compared with the classification rates attained by various other state-of-the-art classification methods published between 2013 and 2021 for the Japanese credit scoring dataset. These results for the single statistical, hybrid statistical (i.e., Preprocessing-based hybrid classifiers), single intelligent, and hybrid intelligent classifiers (i.e., Preprocessing-based hybrid classifiers, Component combination-based hybrid classifiers, Optimization-based hybrid classifiers, and Postprocessing-based hybrid classifiers) are summarized in Table 4 to Table 7, respectively.

According to performances reported in Table 4, the classification rate of the DILR classifier is first compared with the performance of several widely used single statistical

Table 4 Comparison of the DILR with the Single/ Statistical classifiers implemented on the Japanese benchmark

Author/[ref.]	Year	Classifiers	Performance (accuracy) (%)	Sample segmentation
Current Study	2022	DILR	91.58	80/20 training/testing
Liu et al. [26]	2021	LR	85.33	80/20 training/testing
Guo et al. [27]	2019	LDA	85.80	80/20 training/testing
Zhang et al. [28]	2018	KNN	80.00	80/20 training/testing
Tripathi et al. [29]	2018	NB	84.84	90/10 training/testing
Khashei et al. [30]	2013	QDA	84.40	70/30 training/testing

classifiers such as LR, LDA, QDA, KNN, and NB. These results are the most comparable ones to the DILR classifier since the presented model belongs to this category. As can be seen, the DILR classifier can enhance 7.32% the classification rate of the classic LR, one of the most widely used and well-known single statistical models, which only had a classification rate of 85.33%. Additionally, the presented classifier can elevate the classification rate of other single statistical models such as the LDA, QDA, KNN, and NB by 6.74%, 8.51%, 14.48%, and 7.94%, respectively. In this manner, the lowest and highest improvements of the presented model in the category of single statistical models are obtained by LDA and KNN models by 6.74% and 14.48%, respectively. Lastly, single statistical models on average have a classification rate of 84.07% in distinguishing good and bad borrowers, which is 8.93% lower than the performance of the presented DILR. However, the improvement of the presented classifier is not limited to single statistical classifiers. Moreover, based on Table 5, the DILR classifier can enhance 6.12%, and 6.20% the classification rate of some hybrid statistical models including the Multivariate Adaptive Regression Splines + Gabriel Neighbourhood Graph (MARS, GNG)-NB, and k-means-LR classification models. As can be seen, these preprocessing-based hybrid statistical models on average have a classification rate of 86.27%, which is 6.16% lower than the performance of the DILR model.

The classification rates of some single intelligent classifiers, containing the SVM, DT, MLP, RF, and Extreme Learning Machine (ELM) which had been previously implemented on the Japanese dataset are reported in Table 6. As evident, the presented classifier can enhance 7.59% and 5.17% the performance of the SVM and MLP, which are the most popular single intelligent models for classification purposes, and achieve an 85.12% and 87.08 classification rate, respectively. In addition, the proposed DILR classifier can improve 13.10%, 5.75%, and 7.29%, the classification rate of all other single intelligent models such as the DT, RF, and ELM, respectively. The lowest and highest improvements of the proposed DILR model in the category of single intelligent models are obtained by MLP classifier and DT by 5.17% and 13.10%, respectively. Finally, single intelligent models on average have a classification rate of 85.03% in distinguishing good and bad borrowers, which is 7.71% lower than the performance of the DILR classifier. Whereas, the pair-plot of the Japanese credit scoring data set points out the high level of nonlinearity in this dataset. Thus, it is generally expected that the proposed DILR model, as a linear classifier, cannot accurately classify this data set, and nonlinear classifiers, such as intelligent models, may better and accurately model such that data set.

For a more comprehensive assessment, the classification capability of the DILR is also compared by some more complicated hybrid intelligent models with the types of Preprocess-based, Optimization-based, and Component combination-based hybrid classifiers in both forms of intelligent-intelligent and intelligent-statistical combination. The most well-known of these classification models consist of the Multi Grained Augmented-Gradient Boosting Decision Tree (MGA-GBDT), Extreme Gradient Boosting (XGBoost) + LightGBM + GBDT, EELM- Bat algorithm, NB + MLP + PNN + DT + RBF-Weighted voting, LR + RF + SVM + ANN + GBDT, SOM- MLP + CART + LR (weighted voting) models.

As evident from Table 7, the classification rate of the presented classifier is higher than all these hybrid models, except for the XGBoost + LightGBM + GBDT, an optimization and component combination-based hybrid model. The presented classifier can enhance 5.17%, 3.66%, 4.09%, 5.58, and 4.65%, the performance of the MGA-GBDT, EELM-Bat algorithm, NB + MLP + PNN + DT + RBF-Weighted voting, LR + RF + SVM + ANN + GBDT, SOM-MLP + CART + LR (weighted voting) models. Among them, the LR + RF + SVM + ANN + GBDT model with an 86.74% classification rate achieves the lowest performance for the Japanese dataset. Accordingly, the results

Table 5 Comparison of the DILR with the Hybrid/ Statistical classifiers implemented on the Japanese benchmark

Author/[ref.]	Year	Classifiers	Type	Performance (accuracy) (%)	Sample segmentation
Current study	2022	DILR	Single (statistical)	91.58	80/20 training/testing
Ala'raj and Abbod [31]	2016	(MARS + GNG)-NB	Preprocessing (data filtering and feature selection)	86.30	80/20 training/testing
Tsai [32]	2014	k-means- LR	Preprocessing (feature selection)	86.23	80/20 training/testing

Table 6 Comparison of the DILR with the Single/ Intelligent classifiers implemented on the Japanese benchmark

Author/[ref.]	Year	Classifiers	Performance (accuracy) (%)	Sample segmentation
Current study	2022	DILR	91.58	80/20 training/testing
Liu et al. [26]	2021	DT	80.97	80/20 training/testing
Liu et al. [26]	2021	SVM	85.12	80/20 training/testing
Liu et al. [26]	2021	RF	86.60	80/20 training/testing
Tripathi et al. [33]	2020	ELM	85.36	90/10 training/testing
Tsai [32]	2014	MLP	87.08	80/20 training/testing

demonstrate a 5.58% improvement in the classification capability of the DILR compared to this LR + RF + SVM + ANN + GBDT classifier. In this way, the average classification rate of intelligent hybrid models for this data set is 88.47%. Finally, the classification ability of the DILR is improved by 3.52% compared to the average performance of intelligent hybrid classifiers.

Based on the numerical outcomes of this case study, the superiority of the presented discrete learning-based approach is significantly illustrated compared to statistical/intelligent single/hybrid continuous learning-based approaches. The DILR classifier overall surpasses single statistical classifiers with an 8.93% improvement, hybrid statistical classification models with a 6.16% improvement, single intelligent classification models with a 7.71% improvement, and hybrid intelligent classification models with a 3.52% improvement. Although the significant superiority of the DILR classifier over all the single and hybrid statistical approaches may be expected, the superiority of the presented classifier over other intelligent methods illustrates the effect of the proposed learning algorithm matching on the classification rates of machine learning approaches. In addition, the superiority of the presented classifier over hybrid intelligent models demonstrates the high importance of matching the cost function with the goal function in the learning algorithm of classifiers.

Based on empirical results, although hybrid methods with increasing complexity and cost of modeling have used the advantages of different classifiers to elevate the performance of classification, however, the presented classifier with the lower complexity and cost of modeling and solely by utilizing the discrete learning-based process for training and determining the parameter of the model has been successful in achieving desired classification accuracy and generalizability. Therefore, the significance and efficiency of applying the discrete direction-based learning and parameter estimation processes in the classification models, and thus the consistency of this procedure with the goal function of classification, lead to the presented approach outperforming state-of-the-art single/hybrid, linear/nonlinear, and statistical/intelligent classifiers in the literature.

At last, the average classification rate of the presented DILR classifier for all three benchmark datasets of Australian, German, and Japanese credit is 88.07%, which achieved a 7.47% improvement from the 81.95% classification rate of the conventional LR model. Predicting the ability of customers to pay off debts plays a key role in evaluating financial risk and predicting bankruptcy for financial institutions. Accordingly, the use of the proposed classifier model with high discernment in determining good and bad applicants improves the financial organization's cash flow and stability and reduces the possibility of financial losses. Also, the basis for creating efficient plans for identifying and utilizing opportunities and avoiding financial credit risks will be provided.

Table 7 Comparison of the DILR with the Hybrid/ Intelligent classifiers implemented on the Japanese benchmark

Author/[ref.]	Year	Classifiers	Type	Performance (accuracy) (%)	Sample segmentation
Current study	2022	DILR	Single (statistical)	91.58	80/20 training/testing
Liu et al. [26]	2021	MGA-GBDT	Preprocessing (feature selection)	87.08	80/20 training/testing
Zhang et al. [34]	2021	GBDT + XGBoost + lightGBM	Optimization and component combination (intelligent-intelligent)	93.16	80/20 training/testing
Tripathi et al. [33]	2020	EELM- bat algorithm	Optimization	88.35	90/10 training/testing
Tripathi et al. [29]	2018	NB + MLP + DT + PNN + RBF-Weighted voting	Preprocessing (feature selection) and component combination (intelligent-statistical)	87.98	90/10 training/testing
Zhang et al. [28]	2018	LR + RF + SVM + ANN + GBDT	Component combination (intelligent- statistical)	86.74	80/20 training/testing
Tsai [32]	2014	SOM- MLP + CART + LR (weighted voting)	Preprocessing (feature selection) and component combination (intelligent-statistical)	87.51	80/20 training/testing

5 Conclusion

Various approaches have been introduced in the literature for credit scoring purposes. However, despite the apparent differences between the developed approaches, they all utilize continuous distance-based learning procedures. While it is logical that the compatibility between the goal function and the learning procedure plays a crucial role in achieving the highest performance, this aspect has been overlooked in all classifier models developed previously. On this basis, in this study, a new discrete learning-based classification approach is suggested that can be used for decision-making problems in the category of diagnosis, discernment, and detection issues with different characteristics in diverse fields of financial application. Specifically, three well-known credit scoring benchmark datasets have been selected for evaluating the proposed methodology, and in general, a classification rate of 88.07% has been obtained for the proposed DILR compared to an 81.95% classification rate for the classic version. Although in this research, the proposed approach is exemplarily applied to LR, however, the discrete learning-based approach as a viable, logical, and efficient method can be implemented in all kinds of statistical and intelligent classifiers, which can be regarded as future studies. Also, this approach has the potential to be implemented on different data sets in various application domains to obtain superior performance to the conventional models that can be considered as future research suggestions.

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Declarations

Conflict of interest The authors have no conflict of interest.

Ethical Approval The authors confirm that ethical conduct has been respected.

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Mehdi Khashei received a B.Sc. in Applied Mathematics from the Isfahan University in 2002; the M.Sc. and Ph.D. degrees in Systems Engineering and Industrial Engineering from the Isfahan University of Technology in 2005 and 2013, respectively. He is currently the assistant professor at the Department of Industrial and Systems Engineering, Isfahan University of Technology (IUT), Isfahan, Iran. He is the author or co-author of several scientific papers in international journals or communications to conferences with reviewing committees. Main his research interests include Artificial Intelligence; Soft Computing; Time Series Forecasting; Classification and Clustering, Soft Intelligent Decision Making; and Financial Management and Engineering

Sepideh Etemadi received a B.Sc. in Industrial Engineering from the Shahid Bahonar University of Kerman in 2012; an M.Sc. degree in Industrial Engineering from the Ferdowsi University of Mashhad in 2014. She is currently a Ph.D. student in Industrial Engineering at the Isfahan University of Technology (IUT). The main of her research interests include Uncertainty modeling, Artificial Intelligence; Soft Computing; Forecasting; Classification and Clustering, and Financial Management and Engineering

Negar Bakhtiarvand received her B.Sc. and M.Sc. in Industrial Engineering from the Isfahan University of Technology (IUT) in 2016 and 2018, respectively. She is currently a Ph.D. student in Industrial Engineering at the Isfahan University of Technology (IUT). The main of her research interests include Statistical and Intelligent data analysis; Classification and Clustering, and Financial Management and Engineering

Authors and Affiliations

Mehdi Khashei[1,2] · Sepideh Etemadi[1] · Negar Bakhtiarvand[1]

✉ Mehdi Khashei
Khashei@cc.iut.ac.ir

[1] Department of Industrial and Systems Engineering, Isfahan University of Technology (IUT), Isfahan, Iran

[2] Business Intelligence and Knowledge Management (BIKM) Research Group, Isfahan University of Technology (IUT), Isfahan 84156-83111, Iran